

for superconductors with embedded static impurities. However, TLS impurities are dynamical, described by H_{TLS} and the coupling to the electronic system, $H_{\text{e-TLS}}$. In general non-TLS impurities are also present in the superconductor, giving rise to an electron-impurity interaction that can be treated using standard T-matrix methods to describe the static random potential [28]. In the case of conventional isotropic BCS superconductors static non-magnetic impurities do not suppress T_c , modify the equilibrium gap or generate quasiparticle states below the gap [11, 12], but do limit the a.c. conductivity and field penetration into the superconductor, and can be included using standard methods [28]. In what follows we focus on the role of TLS impurities. This model was considered by Maekawa et al. [24]. Our formulation, analysis and results differ significantly from theirs, and other authors as discussed in Sec. V 2. We report our results, with discussion, in Secs. III and V. [29]

The matrix elements for the interaction of electrons and a TLS impurity then reduce to

$$V_{ba}(\mathbf{r}) = \int d^3X \phi_b^*(\mathbf{X}) V(\mathbf{r} - \mathbf{X}) \phi_a(\mathbf{X}), \quad (7)$$

where $a, b \in \{e, g\}$. Thus, the electron-TLS interaction potential, for a single TLS impurity, can be expressed as a 2×2 matrix in the TLS isospin space,

$$V(\mathbf{r}) = \begin{pmatrix} \langle e|V|e\rangle & \langle e|V|g\rangle \\ \langle g|V|e\rangle & \langle g|V|g\rangle \end{pmatrix}, \quad (8)$$

or in terms of the Pauli matrix representation [30]

$$V(\mathbf{r}) = v(\mathbf{r}) \mathbb{1} + m(\mathbf{r}) \sigma_z + n(\mathbf{r}) \sigma_x. \quad (9)$$

The σ_x term is the matrix element for electron scattering by the TLS impurity that is combined with an impurity transition from the ground (excited) to the excited (ground) state. The absence of a σ_y term is because we chose the phases of the wave functions, $\phi_{e,g}(X)$, to be real. The v term, proportional to the 2×2 unit matrix, $\mathbb{1}$, corresponds to the matrix element for the interaction of electrons with a static impurity, i.e. neglecting the level splitting and transition amplitude of the TLS impurity.

Thus, the interaction of electrons with a distribution of TLS impurities is defined by the Hamiltonian,

$$H_{\text{e-TLS}} = \sum_{\alpha} \int d^3r \psi_{\alpha}^{\dagger}(\mathbf{r}) \sum_j \left[v_j(\mathbf{r}) \mathbb{1} + m_j(\mathbf{r}) \sigma_z(j) + n_j(\mathbf{r}) \sigma_x(j) \right] \psi_{\alpha}(\mathbf{r}), \quad (10)$$

or upon transforming the field operators to the momentum-space representation,

$$H_{\text{e-TLS}} = \sum_{\alpha} \frac{1}{V} \sum_{\mathbf{k}', \mathbf{k}} c_{\mathbf{k}'\alpha}^{\dagger} \sum_j [v_j(\mathbf{k}', \mathbf{k}) \mathbb{1} + m_j(\mathbf{k}', \mathbf{k}) \sigma_z(j) + n_j(\mathbf{k}', \mathbf{k}) \sigma_x(j)] c_{\mathbf{k}\alpha}, \quad (11)$$

where $f(\mathbf{k}', \mathbf{k}) \equiv \int d^3r e^{-i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{r}} f(\mathbf{r})$ define the matrix elements for the electron-TLS interactions in Eq. (11). Transformation to the momentum space representation also reduces the mean field BCS Hamiltonian to the standard form for a conventional isotropic superconductor,

$$H_{\text{BCS}} = \sum_{\mathbf{k}} \Psi_{\mathbf{k}}^{\dagger} \left(\xi_{\mathbf{k}} \hat{\tau}_3 + \Delta (i s_y) (-i \hat{\tau}_2) \right) \Psi_{\mathbf{k}}, \quad (12)$$

where $\Psi_{\mathbf{k}}^{\dagger} = (c_{\mathbf{k}\uparrow}^{\dagger}, c_{\mathbf{k}\downarrow}^{\dagger}, c_{-\mathbf{k}\uparrow}, c_{-\mathbf{k}\downarrow})$ is the Nambu spinor for the creation (annihilation) of an electron of momentum $\mathbf{k}$ ($-\mathbf{k}$) and spin $\uparrow$ or $\downarrow$, $\xi_{\mathbf{k}} = k^2/2m - \mu$ is the kinetic energy of electrons with momentum $\pm \mathbf{k}$ relative to the chemical potential, and $\hat{\tau}_{1,2,3}$ denote the 2×2 Pauli matrices in particle-hole space. Note that we have chosen the global phase of the order parameter Δ to be real and positive.

III. TLS ISOSPIN, FERMION REPRESENTATION & STATISTICAL MECHANICS

The isospin operators describing the dynamics of the TLS impurities do not obey the Wick theorem for a perturbation expansion in the framework of quantum field theory. We can circumvent this limitation by introducing local Fermion operators à la Abrikosov that create the ground and excited state of the TLS, i.e. $f_a^{\dagger}|0\rangle = |a\rangle$ where $a \in \{e, g\}$ corresponds to the two levels of the TLS impurity [26].

The dynamics of the TLS isospin operator is then encoded in the local Fermions, “pseudo Fermions” hereafter,

$$\vec{\sigma}(j) = \sum_{a,b} f_{j,a}^{\dagger} \vec{\sigma}_{ab} f_{j,b}, \quad (13)$$

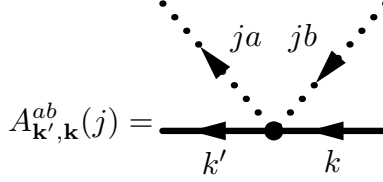


FIG. 2. Feynman diagram for the interaction vertex of conduction electrons and pseudo Fermions.

where $\vec{\sigma}_{ab}$ are the matrix elements of the isospin Pauli matrices. The electron-TLS interaction Hamiltonian can then be written as

$$H_{\text{e-TLS}} = \frac{1}{V} \sum_{\mathbf{k}', \mathbf{k}; \alpha} \sum_j c_{\mathbf{k}'\alpha}^\dagger v_j(\mathbf{k}', \mathbf{k}) c_{\mathbf{k}\alpha} + \frac{1}{V} \sum_{\mathbf{k}', \mathbf{k}; \alpha} \sum_{j;a,b} c_{\mathbf{k}'\alpha}^\dagger f_{j,a}^\dagger A_{\mathbf{k}', \mathbf{k}}^{ab}(j) f_{j,b} c_{\mathbf{k}\alpha} \quad (14)$$

where $v_j(\mathbf{k}', \mathbf{k})$ is the static contribution to the matrix element for quasiparticle scattering by the random potential of TLS impurities, and

$$A_{\mathbf{k}', \mathbf{k}}^{ab}(j) = m_j(\mathbf{k}', \mathbf{k}) (\sigma_z)_{ab} + n_j(\mathbf{k}', \mathbf{k}) (\sigma_x)_{ab}, \quad (15)$$

are the amplitudes that depend on the internal state of the TLS atoms encoded by the pseudo-Fermion operators and the Pauli matrix elements for each TLS impurity located at the random sites labelled by index j . This vertex is represented by the Feynman diagram in Fig. 2. However, the transformation defined by Eq. (13) introduces unphysical states, i.e. the empty state $|0\rangle$ and the doubly occupied state $|eg\rangle$, both of which must be excluded in carrying out ensemble averages of physical operators [26].

A. Popov-Fedotov Projection

The Nambu Green's function in the Matsubara formalism for a superconductor with an embedded distribution of TLS impurities is given by

$$\hat{G}(x_1, x_2) = -\langle T_\tau \Psi(x_1) \Psi^\dagger(x_2) \rangle, \quad (16)$$

where $\Psi(x) = \text{col}(\psi_\uparrow(\mathbf{r}), \psi_\downarrow(\mathbf{r}), \psi_\uparrow^\dagger(\mathbf{r}), \psi_\downarrow^\dagger(\mathbf{r}))$ is the 4-component Fermion Nambu spinor, and the ensemble average is taken over the grand canonical ensemble defined by the Hamiltonian,

$$H = H_{\text{BCS}} + H_{\text{TLS}} + H_{\text{e-TLS}}, \quad (17)$$

with Eqs. (12), (4), and (14). The average denoted by $\langle \dots \rangle$ in Eq. (16) is a constrained ensemble average over pseudo Fermion states with occupancy $n_f = 1$ corresponding to the two physical states $|e\rangle$ and $|g\rangle$ for any of the TLS impurities,

$$\langle \mathcal{O} \rangle \equiv e^{\beta\Omega} \text{Tr} \left\{ e^{-\beta H} \mathcal{O} \right\} \Big|_{n_f=1}. \quad (18)$$

The partition function, and corresponding thermodynamic potential, are also calculated in the physical subspace,

$$e^{\beta\Omega} = 1 / \text{Tr} \left\{ e^{-\beta H} \right\} \Big|_{n_f=1}. \quad (19)$$

As formulated the constraint is incompatible with finite-temperature quantum field theory where ensemble averages are performed over the entire Hilbert space. In order to circumvent this problem and still enforce the constraint we use the Popov-Fedotov method. Note that pseudo-Fermion number, $\sum_a f_a^\dagger f_a$, is conserved by the Hamiltonian $H_{\text{e-TLS}}$. Thus, following Popov and Fedotov, we can implement the constrained statistical average as an unconstrained trace over states of conduction electrons and the pseudo-Fermion Fock space by introducing a pseudo-Fermion chemical potential,

$$\langle \dots \rangle = e^{\beta\Omega} \text{Tr} \left\{ e^{-\beta[H - \mu_f(n_f - 1)]} \dots \right\}. \quad (20)$$

The physical constraint $n_f = 1$ is achieved by choosing $\mu_f = -i\pi T/2$, which enforces zero statistical weight to the pseudo-Fermion Fock states with $n_f = 0$ and $n_f = 2$, thus eliminating the unphysical states from the

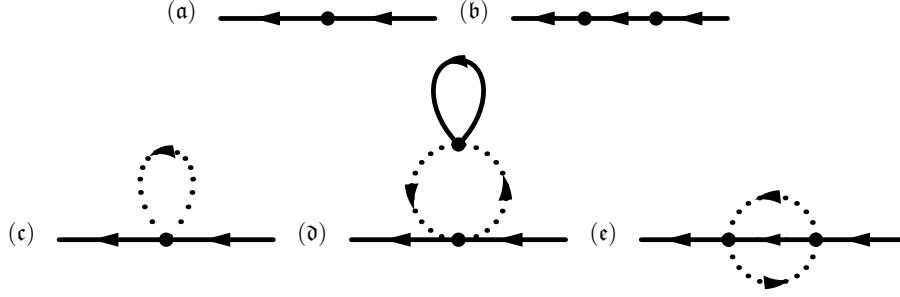


FIG. 3. Leading order diagrams for the quasiparticle-TLS impurity scattering corrections to the Green's function. Solid lines represent the Gorkov propagator for quasiparticles and pairs, dotted lines represent the propagator for pseudo Fermions, solid circles represent the interaction vertex between Fermions and TLS impurity potential. Diagrams (a) and (b) represent the interaction with the static component of the random potential generated by the distribution of impurities. Diagram (c) is the 1st order contribution from pseudo-Fermions and the random potential, while diagrams (d) and (e) encode intermediate states of the TLS impurities.

ensemble average [27]. Specifically, the terms introduced by the chemical potential $e^{\beta\mu_f(n_f-1)}$ have no effect on the trace over physical states $|n_f = 1\rangle$, but introduce a factor of $+i(-i)$ for the unphysical states $|n_f = 0\rangle$ ($|n_f = 2\rangle$). These phase factors enforce cancellation of the contributions to the ensemble average from the Fock states with $n_f = 0$ or $n_f = 2$. The key result is that Abrikosov's pseudo-Fermion representation of the TLS isospin with the Popov-Fedotov projection of the unphysical pseudo-Fermion states allows us to employ traditional quantum field theory techniques, including perturbation theory based on the Wick theorem and the corresponding Feynman rules.

For example, we can now calculate the unperturbed pseudo-Fermion propagator,

$$D_{i,a;j,b}(\tau - \tau') \equiv - \left\langle T_\tau f_{i,a}(\tau) f_{j,b}^\dagger(\tau') \right\rangle_0, \quad (21)$$

where the subscript i,a denotes the a -state of the pseudo-Fermion field operator of the i^{th} impurity. In the Matsubara formalism we have

$$D_{i,a;j,b}(\varepsilon_n) = \frac{\delta_{ij}\delta_{ab}}{i\varepsilon_n - \varepsilon_{j,a} + \mu_f} = \frac{\delta_{ij}\delta_{ab}}{i\tilde{\varepsilon}_n - \varepsilon_{j,a}}, \quad (22)$$

where $\tilde{\varepsilon}_n = (2n + \frac{1}{2})\pi T$ is the shifted Matsubara energy, and $\varepsilon_{j,\pm} = \pm E_j/2$ are the two energy levels of the j^{th} TLS impurity.

The presence of two-level tunneling centers enlarges the Hilbert space to include the quantum states of the random distribution of TLS impurities in the ensemble average. The statistical averaging is implemented by introducing pseudo-Fermions and the Popov-Fedotov constraint to represent the quantum states and dynamics of the TLS impurities as described above. In particular, we can formulate Feynman perturbation theory to calculate the corrections to the quasiparticle and Cooper pair propagators. Figure 3 shows the Feynman diagrams for the corrections to the Nambu matrix propagator through second-order in the random potential of the TLS impurity distribution.

The solid arrowed lines represent the Gorkov propagator for quasiparticles and pairs, while the dotted lines with arrows represent the propagator for pseudo Fermions. The solid circles represent the interaction vertex between Fermions/pairs and random potential. The first two diagrams, (a) and (b), represent the interaction with the *static* component of the random TLS potential. The other three diagrams, (c), (d) and (e), describe intermediate quantum states, mediated by the pseudo-Fermions, of the random TLS potential.

It is important to note that Maekawa et al. [24] developed a field theory approach to the role of TLS disorder on superconductivity, adopting Abrikosov's method [26] to eliminate the fictitious pseudo-Fermion states $|0\rangle$ and $|eg\rangle$. However, this method fails to obey the linked-cluster theorem [31], and thus includes unphysical diagrams in the self energy. Implementing the Popov-Fedotov projection method avoids this error.